Mukhyansh

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Education

Graphic Era Hill University Uttarakhand
Bachelor of Technology in Computer Science & Engineering 2024 – Present ***CGPA: 8.50***

Siddhartha Public School Delhi
Class XII 2022-2024 ***Score: 81.6%***

Technical Skills

Languages: C, C++, Java, JavaScript, SQL
BackEnd / Databases: Node.js, MySQL, PostgreSQL
FrontEnd: HTML, CSS
Tools: Git, GitHub, Docker, VSCode, Linux
Coursework: Data Structures & Algorithms, Object Oriented Programming, Operating Systems

Projects

Adaptive User-space Scheduler (C, Python, LinuxOS, PostgreSQL)

August 2026 – Present

- Built an adaptive CPU scheduling system for LinuxOS that catches the process telemetry including CPU usage, I/O workload metrics, Context Switches etc.
- The system is designed such that it collects and stores data in a database, trains Models based on that data, and dynamically adjusts scheduling priorities based on predicted CPU ,I/O and Memory needs.
- Developed a closed loop like architecture that stores results from the predicted run and further trains the models based on those results for improvement and maximum accuracy.
- Tested using both Synthetic processes and actual running processes on the OS against linux traditional EEVDF and CFS scheduling algorithms.

Distributed File Transfer System (*C, Compression Algorithms, VSCode, TCP Sockets)*

Jan 2026 – March 2026

- Built a distributed file transfer system with self-implemented file compressor using LZW Compression Algorithm and networking concepts.
- Structured into 3 major modules: Connectivity, Compression and File Handling.
- Successfully implemented connectivity between 2 devices on the same LAN or localhost.
- Achieved 60% reduction in file size for typical text based files before transmission.

Thread Library (*C, VSCode, Context Switching, Multithreading)*

Dec 2025 – March 2026

- Implemented a Pre-emptive Thread Library in C language that provided User level Thread Management with custom TCB (Thread Control Block).
- The library itself contains 8+ core threading functionalities like signals, ucontext and setitimer() etc.
- Implemented a process scheduler from scratch following the Round Robin scheduler format.
- Achieved a 35% improvement in runtime in frequent context-switching scenarios against the POSIX pthread library.

Achievements

- Solved 450+ problems in LeetCode and CodeForces.
- 1700+ rated in contests in LeetCode.
- Led the Team in Smart India Hackathon and got selected out of 300 teams.
- Got Second place in a University Level Debate Competition (Debathon).